**THE CATHOLIC DIOCESE OF TANGA**
**TANGA CATHOLIC SECONDARY SCHOOLS UNION (TACASSU)**
**FORM FOUR JOINT PRE-NATIONAL EXAMINATION 1**

**032/2A CHEMISTRY 2A**
**(ACTUAL PRACTICAL)**
**MARKING SCHEME 2026**

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**QUESTION 1**: **(25 MARKS)**
Reaction:
$CH_3COOH+NaOH \rightarrow CH_3COONa+H_2O$

(vi) Table of Results **(8 Marks)**
Pipette 20.0 $cm^3$

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| **Experiment** | **Pilot** | **1** | **2** | **3** |
|---|---|---|---|---|
| Final Reading (cm3) | 20.50 | 20.00 | 19.90 | 20.10 |
| Initial Reading (cm3) | 00.00 | 00.00 | 00.00 | 00.00 |
| Volume used (cm3) | 20.50 | 20.00 | 19.90 | 20.10 |

(a) Colour change: From pink to colourless **(2 Marks)**

(b) $\frac{20.00 + 19.90 + 20.10}{3} = 20.00\ cm^3$

(c) 20 $cm^3$ of solution **BB** required 20 $cm^3$ of solution **AA** for complete reaction.

(d) Concentration of Acetic Acid in $mol/dm^3$ **(4 Marks)**

Using:
$C_1V_1=C_2V_2$
Substitution:
$0.1 \times 20 = C \times 20$
Therefore:
$C=0.10\ mol\ dm^{-3}$

(e) Concentration in $g/dm^3$ **(3 Marks)**

Molar mass:
$(2\times12)+(4\times1)+(2\times16)=60$
Calculation:
$0.10\times60=6\ g\ dm^{-3}$

(f) Molarity of Commercial Vinegar Before Dilution **(4 Marks)**

Using dilution formula:
$C_1V_1=C_2V_2$
Substitution:
$C_1\times1=0.10\times8.4$
Therefore:
$C_1=0.84$ mol $dm^{-3}$

**QUESTION 2**: **(25 MARKS)**

(a)

(i) Completed Table **(5 Marks)**

| Exp. | Volume of AB $Cm^3$ | Volume of CD $Cm^3$ | Temperature $^0C$ | Time(t) sec | 1/t ($sec^{-1}$) |
|---|---|---|---|---|---|
| 1 | 10 | 10 | 80 | 5.0 | 0.20 |
| 2 | 10 | 10 | 70 | 7.5 | 0.13 |
| 3 | 10 | 10 | 60 | 12.5 | 0.08 |
| 4 | 10 | 10 | 50 | 40.0 | 0.025 |

(ii) Aim of Experiment **(2 Marks)**
To investigate the effect of temperature on the rate of chemical reaction.

(iii) Redox Reaction **(2 Marks)**
A reaction in which oxidation and reduction occur simultaneously.

(iv) Substance Reduced **(2 Marks)**
Potassium permanganate was reduced.
Oxidation state of Mn changed from +7 to +2.

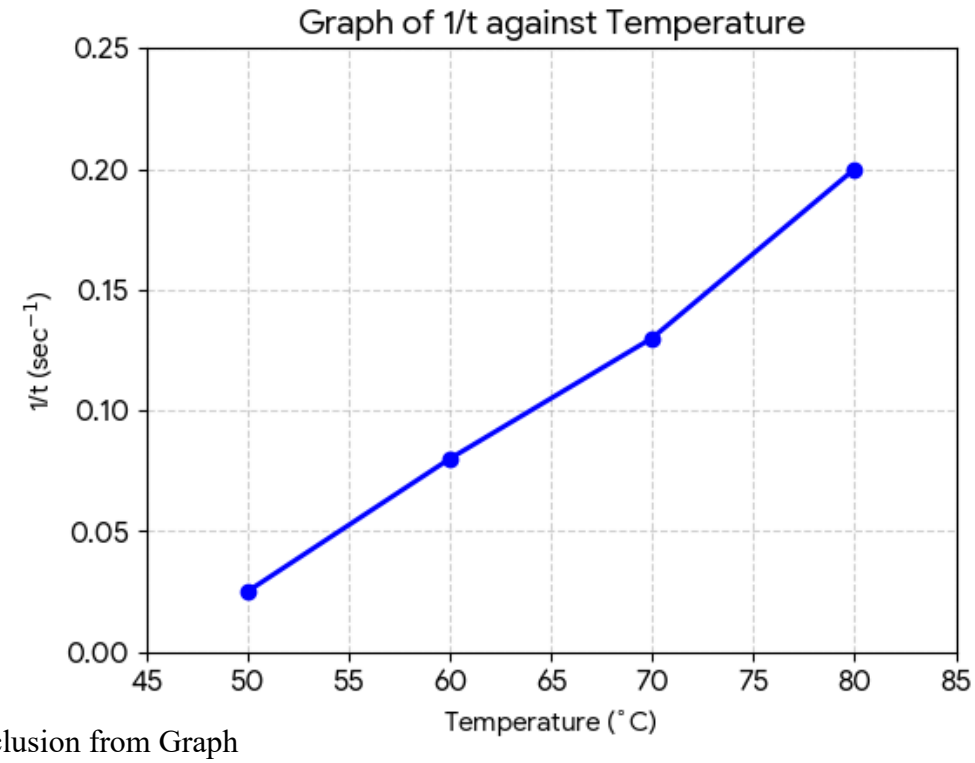
(v) Property Used to Measure Rate: Disappearance of purple colour. **(2 Marks)**

(b)

(i) Meaning of 1/t:Relative rate of reaction. **(1 Mark)**

(ii) Factor Affecting Rate: temperature **(1 Mark)**

(iii) Graph of 1/t Against Temperature **(7 Marks)**




(c) Conclusion from Graph

(i) Increase in temperature increases reaction rate.

(ii) Refrigeration lowers reaction rate.

(iii) Refrigeration slows milk spoilage.

(**Any three points × 1 mark = 3 Marks**))

(d) Test for Gas Produced **(2 Marks)**

Gas is carbon dioxide.

Equation:

$Ca(OH)_2\ (aq) + CO_2\ (g) \rightarrow CaCO_3\ (s) + H_2O\ (l)$

Observation:

Limewater turns milky.